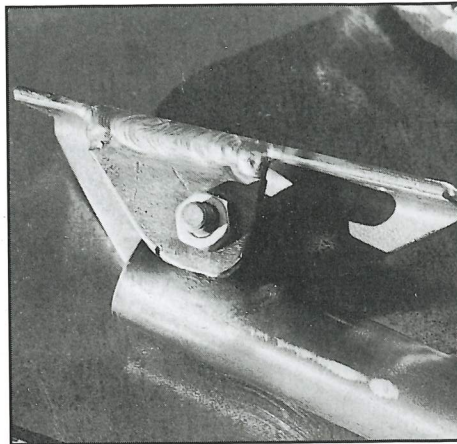
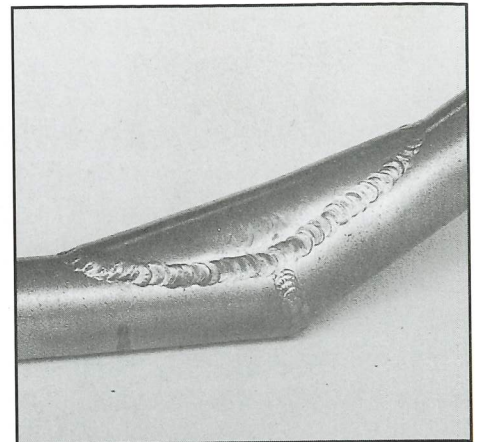




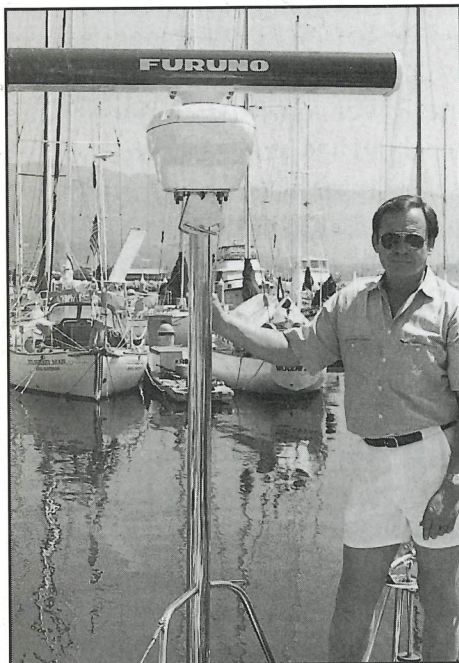
Troubleshooting defective welds is how you learn to be a better welder. Rough welds and smoky deposit at the weld indicates not enough argon flow, or in certain cases, it could indicate contaminated argon, or even the wrong gas in the bottle.



Another defective weld. Note the small crater at the right side of the weld bead, caused by shutting the torch off abruptly while the puddle was still molten.



This otherwise excellent TIG weld on this aircraft 4130 steel tube mount is defective because the weld bead is concave and subsequently weak in one spot. Adding another weld bead on top of the low one would be acceptable.



I welded this boat radar antenna mount from 316 stainless steel. 316 grade stainless is least susceptible to salt water corrosion. The radar antenna manually tilts to match the tilt of the yacht when under full sail. Photo: "Vicky."

START, if so equipped.

- Adjust gas flow to 20 cfh.
- Set water-cooling switch to ON, if so equipped.

•Set coarse- and fine-amp controls—coarse to 75 amps, fine to 80% for 0.063" steel. This gives $75 \times 0.80 = 60$ amps with the pedal all the way down. For thick metal, try these settings: coarse to 125 amps and fine to 35%, or 44 amps with full pedal. If this doesn't provide enough heat, increase the fine-amp setting. Go 50%, 80% or whatever heat you need. The reason for the lower fine-amp setting for thicker materials is to increase duty cycle. Regardless of the fine-amp setting, final amperage at the electrode is controlled by you at the pedal.

•Turn on contactor and amp-control switches to remote foot pedal, if so equipped. Contactor switch turns "on" the foot pedal.

TIG WELDING STAINLESS STEEL

TIG welding stainless steel is similar to welding mild steel and 4130 steel. However, you'll need some extra items: an extra bottle of inert gas, extra flow meter and about 20

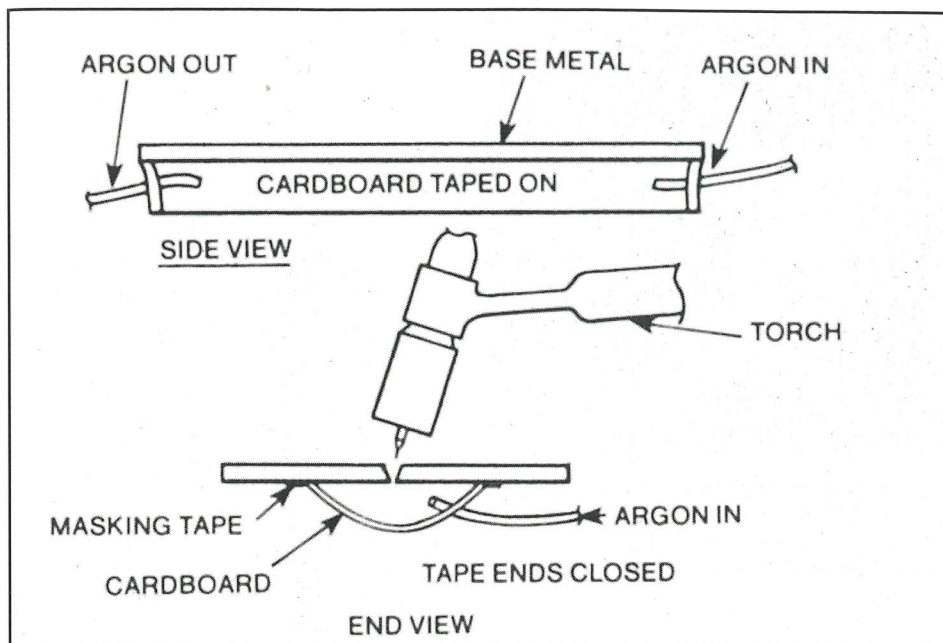
feet of argon hose. This assembly of items will be used to purge, or shield, the back side of the weld.

Back-Gas Purging

As mentioned earlier, purging is the process of displacing all atmospheric gases and replacing them with an inert shielding gas such as argon. I usually set purge-gas flow the same as or about 25% more than torch flow.

The reason for using back-gas purge is that molten stainless crystallizes if it's exposed to air. Sugar, or crystallization, on the back side of the weld would weaken the weld and base metal considerably.

It's easy to back-gas stainless tubing such as an engine exhaust pipe. You simply fill the tube with gas. Cap both ends of the tube with making tape, punch holes in the tape, stick the purge hose in one end and a short piece of open tubing in the other end to exhaust the argon. At 15-20 cfh gas flow, the average exhaust pipe can be purged in air in 4-5 minutes. Larger pieces, such as long, 12" ID tubing, take more time to purge.



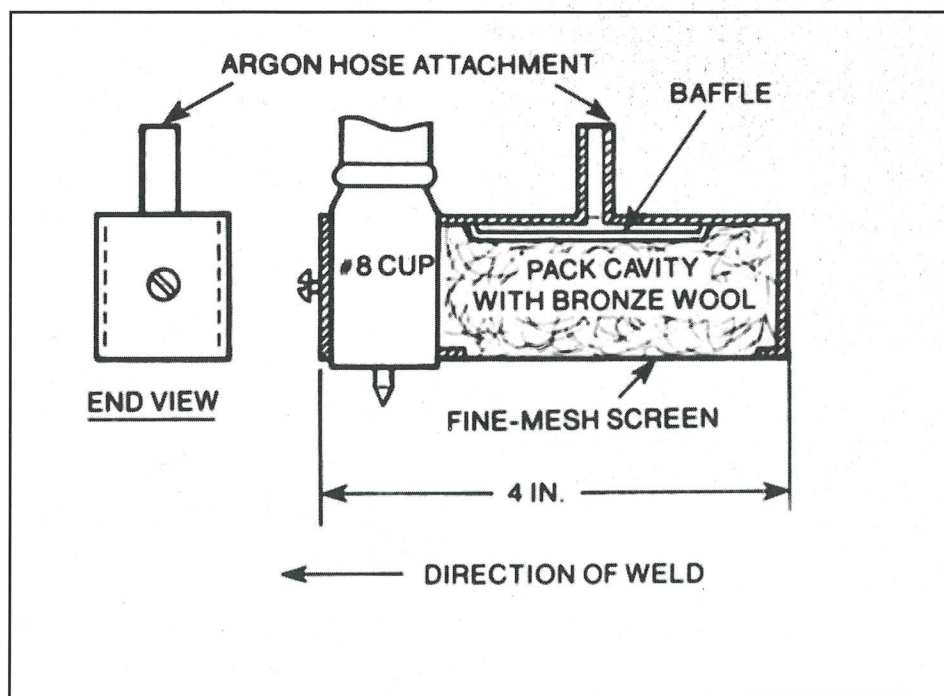
Simple back-gas shield for TIG welding titanium can be made from cardboard. Masking tape is used to hold it to workpiece. Use the same kind of shield for welding stainless steel and 4130 steel.

Flat stainless plate is harder to purge. But you can build a cardboard-and-mask-tape cover over most seams to act as a purge shield. Simply enclose the back side of the weld seam with craft-type cardboard and masking tape. Shape the cardboard so it straddles, but doesn't touch, the weld seam. Burn a large hole in the cardboard and the purge gas will be lost. Cap the ends with masking tape and insert the purge and exhaust hoses. The accompanying drawing illustrates how to make a back-gas shield for flat plate.

TIG WELDING TITANIUM

Although most alloys of this expensive, lightweight metal can be TIG welded with the same basic machine setup as mild steel and 4130 steel, it requires a more elaborate shielding-gas apparatus than stainless steel. As a result, some titanium alloys are not weldable at all. This is because

of titanium's extreme sensitivity to contamination. Hot titanium reacts with the atmosphere and dissimilar metals, causing weld embrittlement.



Trailing-gas shield for TIG welding titanium protects completed welds as it cools. Shield must fit tightly to cup. Practice on stainless steel before welding expensive titanium parts.

This contamination is serious if carbon, oxygen or nitrogen is present in sufficient quantities. In the solid state, as in a weld heated above 1200° F (649C), titanium absorbs oxygen and nitrogen from the air.

An argon-gas shield must cover both sides of a weld seam while titanium is at the 3263° F (1795C) molten stage and all the way down to 800° F (427C). Otherwise, embrittlement from contamination and resulting cracking will occur.

Make a device to provide a trailing shield to cover the weld until it cools to below 800° F (427C). Usually, after the trailing shield passes over the completed weld, the titanium has cooled sufficiently. You can also use temperature-indicating crayon or paint to be sure. But if you do, be careful not to contaminate the weld with the crayon or paint. See the drawings of trailing-shield device.

Gas Chamber

Because it is extremely reactive with nitrogen, oxygen and hydrogen, the best place to weld titanium is in a total inert atmosphere such as that in outer space. So, if you can get a ride on the Space Shuttle to do your welding, great. Otherwise, the next best thing is a gas chamber. Such a chamber looks similar to an incubator for newborn babies or a bead-blasting cabinet. Put the part to be welded inside the chamber, close it, then put your hands into built-in gloves to do the welding. The chamber is completely purged with argon so no air exists. Much purer titanium welds will result.

TIG WELDING ALUMINUM

TIG welding aluminum is slightly different than welding steel. Machine settings are different. And, as you may remember from the gas-welding chapter, aluminum doesn't change color as it forms a puddle—it gets shiny instead. However, unlike gas welding, flux isn't needed to TIG weld aluminum. In fact, flux would really mess up the weld and your TIG welder. The following tips will help you TIG weld aluminum:

- The weld seam area should be as clean as possible, and it should be free of aluminum oxide. Remove the oxide immediately before you weld—not a week or two before—by mechanical or chemical cleaning. Mechanical cleaning is done with a stainless steel wire brush, sandpaper or abrasive pads. Afterward, wash off the dust with soapy water and rinse with clear water. Then, wipe down the seam area with alcohol, MEK

or acetone.

- To eliminate the need for cleaning newly fabricated aluminum parts, use the more expensive paper-covered sheet stock. Adhesive-backed, paper-covered sheet can be cut, formed and fitted with the covering in place. When you're ready to weld, simply peel back the covering from the weld seam to expose super-clean aluminum, and you're ready to weld.

- If TIG welding aluminum castings or forgings, V-groove the joint all the way through. Or V-groove the joint from both sides and weld on both sides, if possible.

KNOW BEFORE YOU START

Because welding titanium is so specialized and the technique so critical, you should discuss your titanium-welding project with a welder who specializes in titanium before attempting it. If you can't find such a person, RMI Company, Niles, OH 44446 is a company that specializes in welding titanium. They would be glad to answer your questions. Or contact the American Society for Metals, Metals Park, OH 44073. Their educational division, the Metals Engineering Institute, can provide instructional material through their Home Study and Extension Courses for all types of welding techniques and metals. Finally, there's the American Welding Society (AWS), 2501 Northwest 7th Street, Miami, FL 33125. Their four-volume *Welding Handbook* will give you all you ever wanted to know about welding and more.

GAS SHIELDING FOR TITANIUM WELDING

Material Thickness	Torch Argon Flow (cfh)	Trailing Shield Argon Flow (cfh)	Back Gas Helium or Argon (cfh)
0.030	15	15	3
0.060	15	20	4
0.090	20	20	4
0.125	20	30	5

- If welding aluminum plate that's more than 1/16" thick, V-groove the joint for better penetration.

- Don't attempt to weld 2024 aluminum or other non-weldable aluminum alloys. Read on for a handy test to check the weldability of unknown aluminum alloys. Weldable alloys include 1100, 5052, 6061 and all castings.

Machine Settings

- Polarity selector switched to AC.

- High-frequency unit set to CONTINUOUS—an absolute necessity.